

ECON403 Introduction to Contract Theory and Auctions

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

November 17, 2023

What are auctions?

- Auctions are mechanisms for the buying and selling of objects by way of bids submitted by potential buyers.
- They are known since antiquity.
- Spectrum licensing, mineral and oil rights, clean energy procurement, utilities procurement, internet advertisements, art, fresh fish.
- The 2020 Nobel Prize in Economics was awarded to Bob Wilson and Paul Milgrom for having “improved auction theory and invented new auction formats, benefiting sellers, buyers and taxpayers around the world”.

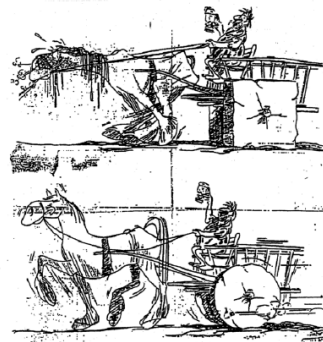
- Official Prerequisites: ECON201 and ECON301 with C- or above.
- Unofficial Prerequisites: Game Theory, ECON302.
- The class is actually a good continuation for Economics of Information, ECON335 and/or Advanced Microeconomics ECON305.
- Another unofficial prerequisite: good background in Mathematics, since you will be able to enjoy and understand the course much more.

The course itself

- Topics:
 - Advanced Introduction to Game Theory: Games of Incomplete Information.
 - The classical auction formats under complete information.
 - Bayesian Auctions: classical results and extensions.
 - Advanced topics, but depends on class performance/interests.
- Why learn auction theory?
 - The fundamental building-blocks helps understand more complex environments.
 - Tools provide connections to other branches of “standard” economic theory.
 - Develop intuition and insights that can inform other areas in economics.

Why Every Economist Should Learn Some Auction Theory

Paul Klemperer, August 2001.



Auction Theory **Heineken.**
Refreshes the parts other
economics **beers cannot reach.**